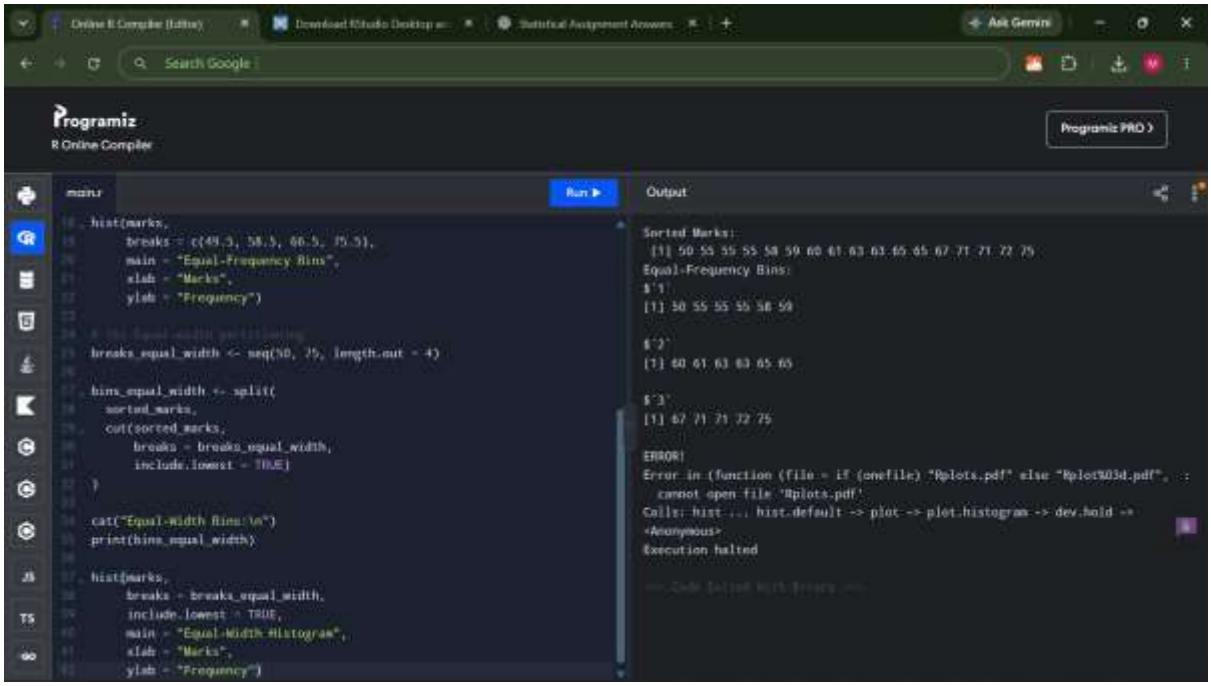
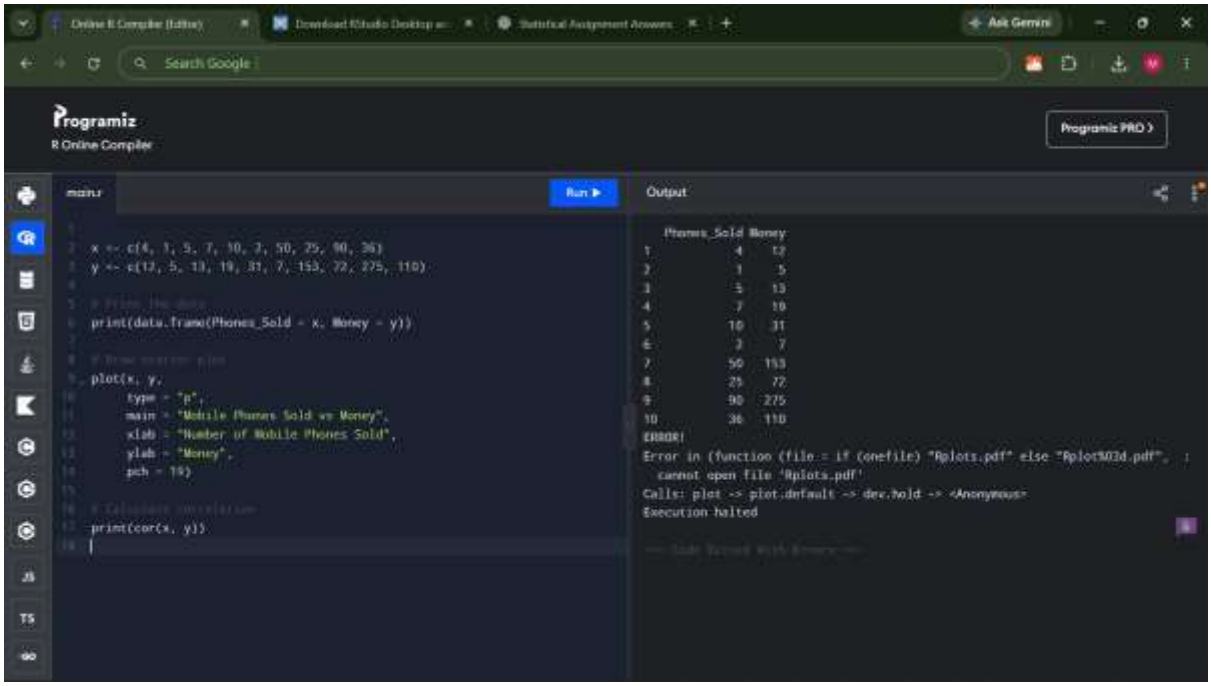
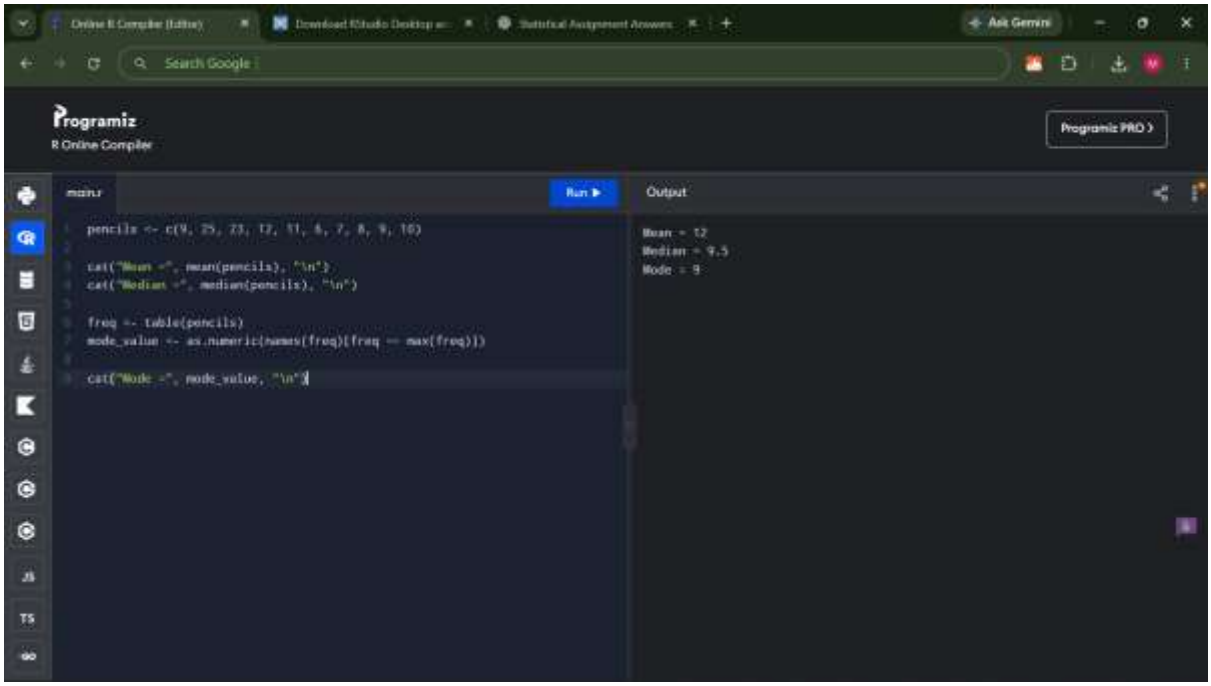
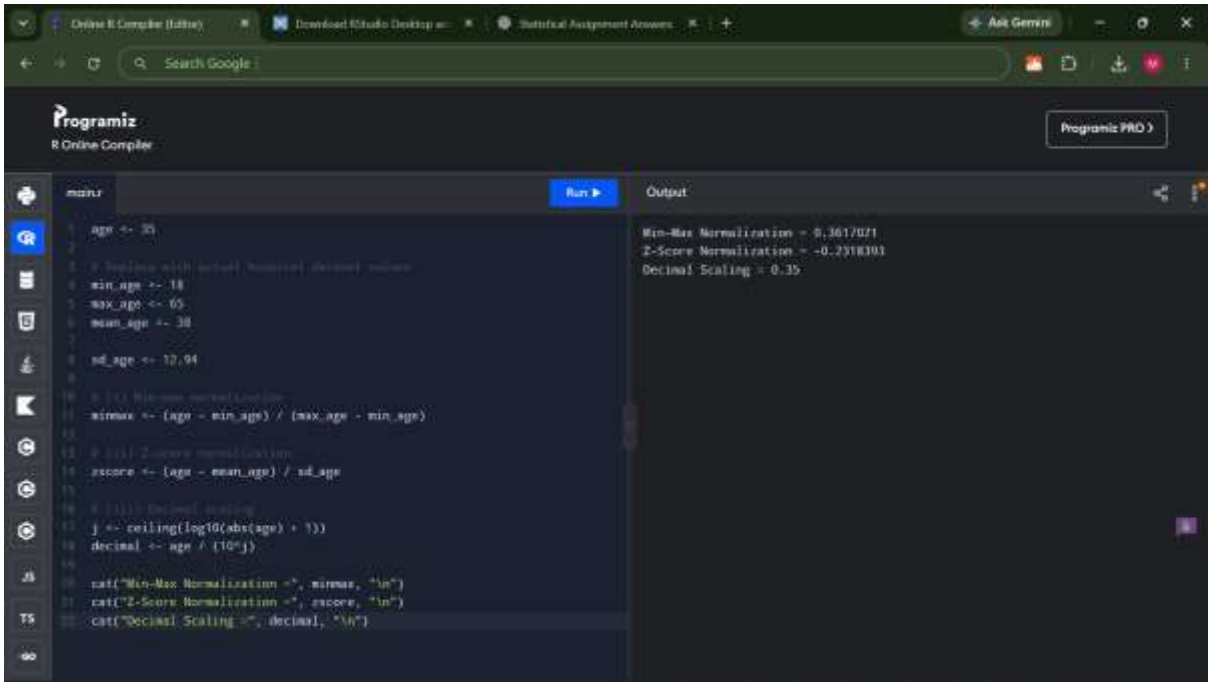
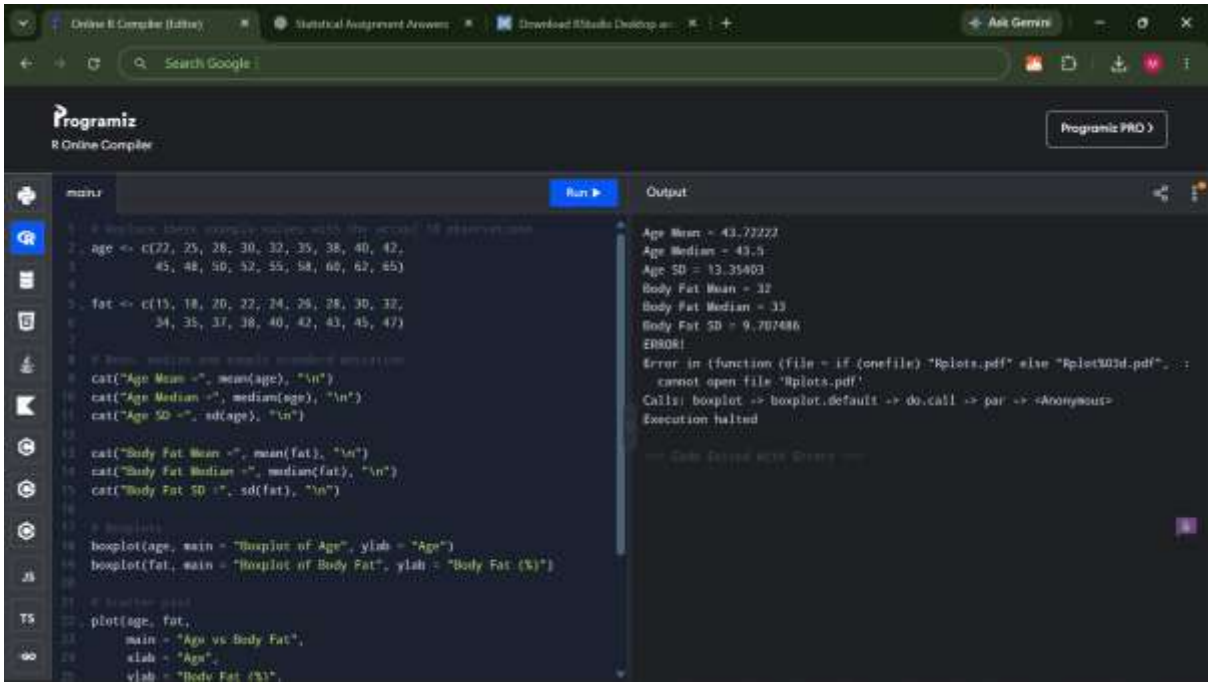
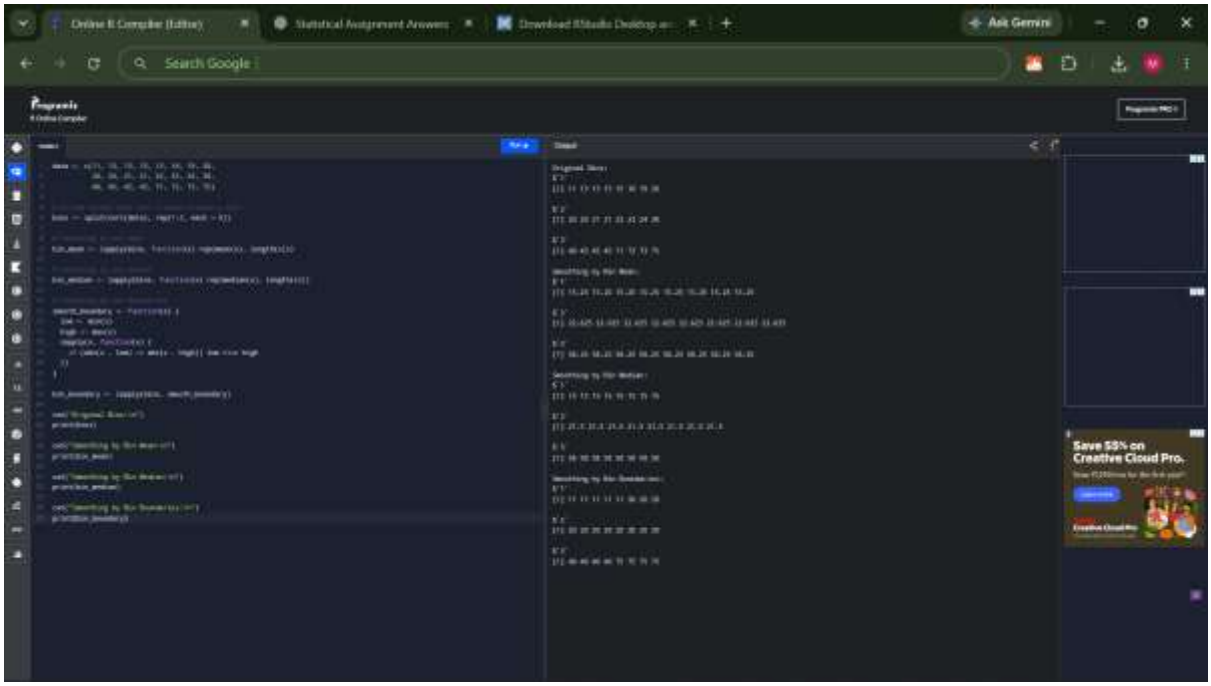

Programiz
R Online Compiler
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main.r
Run
Output
hist(marks,
breaks = c(49.5, 58.5, 66.5, 75.5),
main = "Equal-Frequency Bins",
xlab = "Marks",
ylab = "Frequency")
breaks_equal_width <- seq(50, 75, length.out = 4)
bins_equal_width <- split(
sorted_marks,
cut(sorted_marks,
breaks = breaks_equal_width,
include.lowest = TRUE)
)
cat("Equal-Width Bins:\n")
print(bins_equal_width)
hist(marks,
breaks = breaks_equal_width,
include.lowest = TRUE,
main = "Equal-Width Histogram",
xlab = "Marks",
ylab = "Frequency")
Sorted Marks:
[1] 50 55 55 55 58 59 60 61 63 63 65 65 67 71 71 72 75
Equal-Frequency Bins:
$`1`
[1] 50 55 55 55 58 59
$`2`
[1] 60 61 63 63 65 65
$`3`
[1] 67 71 71 72 75
ERROR!
Error in (function (file = if (onefile) "Rplots.pdf" else "Rplot%03d.pdf",
cannot open file 'Rplots.pdf'
Calls: hist ... hist.default -> plot -> plot.histogram -> dev.hold ->
<Anonymous>
Execution halted

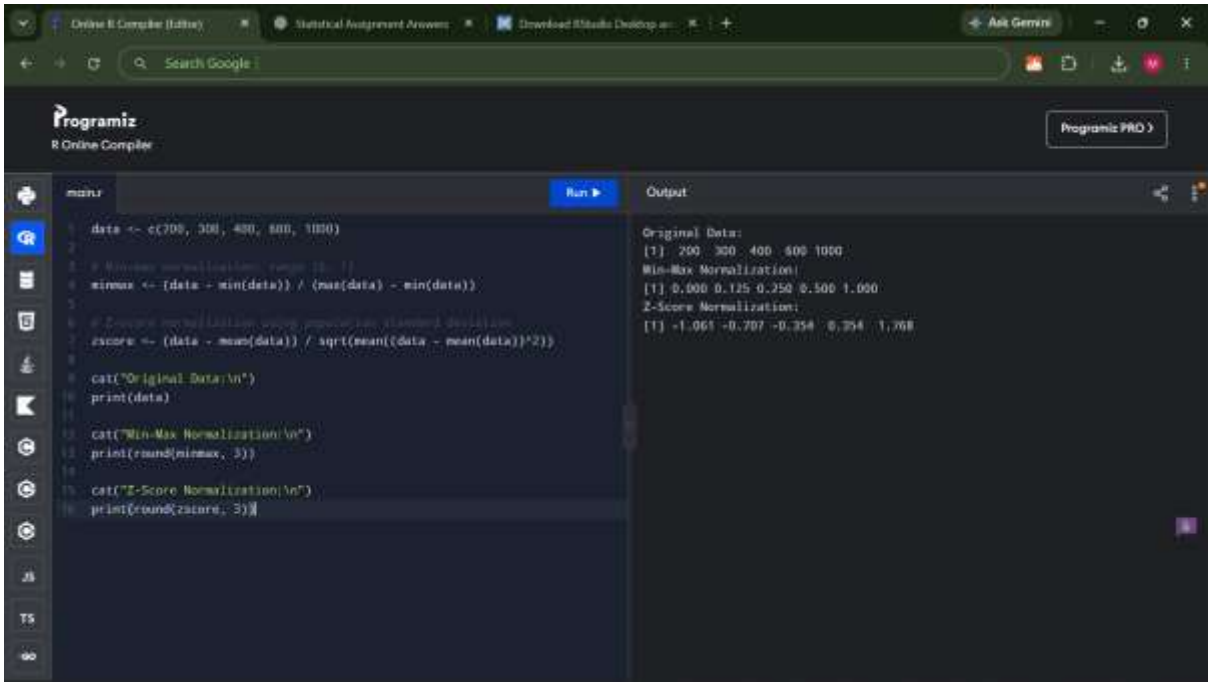

Programiz
R Online Compiler
Programiz PRO
main.r
Run
Output
x <- c(4, 1, 5, 7, 10, 3, 50, 25, 90, 36)
y <- c(12, 5, 13, 19, 31, 7, 153, 72, 275, 110)
print(data.frame(Phones_Sold = x, Money = y))
plot(x, y,
type = "p",
main = "Mobile Phones Sold vs Money",
xlab = "Number of Mobile Phones Sold",
ylab = "Money",
pch = 19)
print(cor(x, y))
Phones_Sold Money
1 4 12
2 1 5
3 5 13
4 7 19
5 10 31
6 3 7
7 50 153
8 25 72
9 90 275
10 36 110
ERROR!
Error in (function (file = if (onefile) "Rplots.pdf" else "Rplot%03d.pdf",
cannot open file 'Rplots.pdf'
Calls: plot -> plot.default -> dev.hold -> <Anonymous>
Execution halted

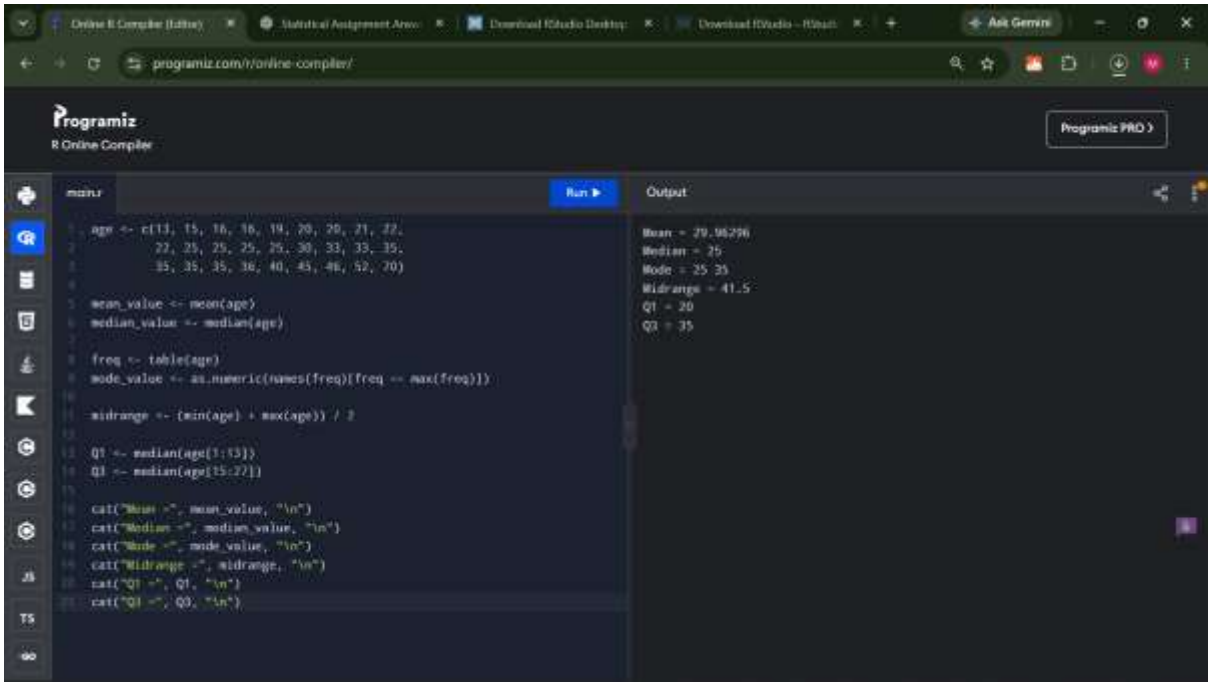

Programiz
R Online Compiler
Programiz PRO
main.r
Run
Output
pencils <- c(9, 25, 23, 12, 51, 6, 7, 8, 9, 10)
cat("Mean =", mean(pencils), "\n")
cat("Median =", median(pencils), "\n")
freq <- table(pencils)
mode_value <- as.numeric(names(freq)[freq == max(freq)])
cat("Mode =", mode_value, "\n")
Mean = 12
Median = 9.5
Mode = 9

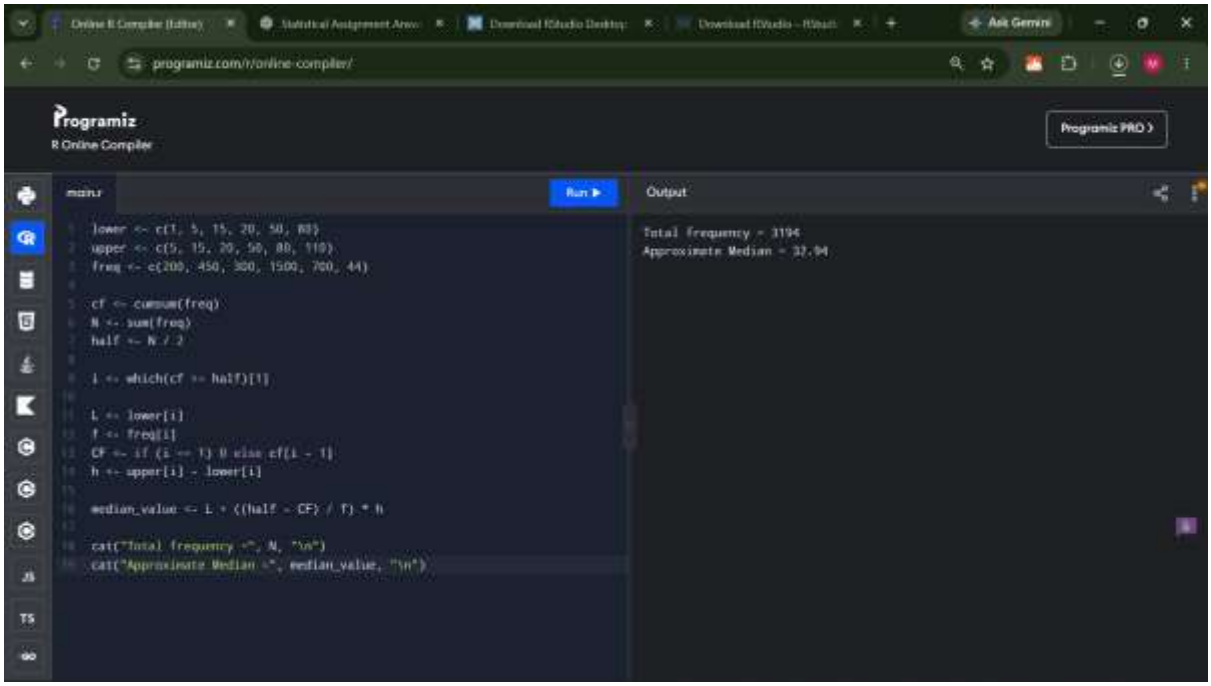

Programiz
R Online Compiler
Programiz PRO
main.r
Run
Output
age <- 35
min_age <- 18
max_age <- 65
mean_age <- 38
sd_age <- 12.94
minmax <- (age - min_age) / (max_age - min_age)
zscore <- (age - mean_age) / sd_age
j <- ceiling(log10(abs(age) + 1))
decimal <- age / (10^j)
cat("Min-Max Normalization =", minmax, "\n")
cat("Z-Score Normalization =", zscore, "\n")
cat("Decimal Scaling =", decimal, "\n")
Min-Max Normalization = 0.3617021
Z-Score Normalization = -0.2318393
Decimal Scaling = 0.35

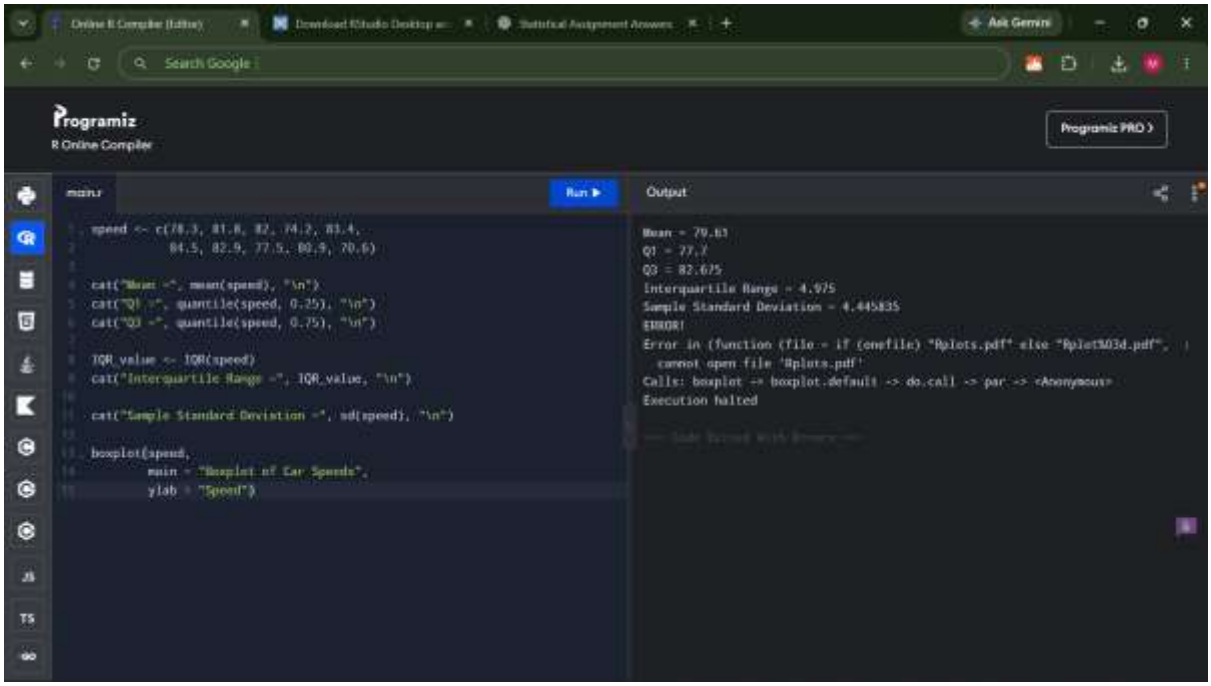

Programiz
R Online Compiler
Programiz PRO
main.r
Run
Output
age <- c(22, 25, 28, 30, 32, 35, 38, 40, 42,
45, 48, 50, 52, 55, 58, 60, 62, 65)
fat <- c(15, 18, 20, 22, 24, 26, 28, 30, 32,
34, 35, 37, 38, 40, 42, 43, 45, 47)
cat("Age Mean =", mean(age), "\n")
cat("Age Median =", median(age), "\n")
cat("Age SD =", sd(age), "\n")
cat("Body Fat Mean =", mean(fat), "\n")
cat("Body Fat Median =", median(fat), "\n")
cat("Body Fat SD =", sd(fat), "\n")
boxplot(age, main = "Boxplot of Age", ylab = "Age")
boxplot(fat, main = "Boxplot of Body Fat", ylab = "Body Fat (%)")
plot(age, fat,
main = "Age vs Body Fat",
xlab = "Age",
ylab = "Body Fat (%)"
Age Mean = 43.72222
Age Median = 43.5
Age SD = 13.35403
Body Fat Mean = 32
Body Fat Median = 33
Body Fat SD = 9.707486
ERROR!
Error in (function (file = if (onefile) "Rplots.pdf" else "Rplot%03d.pdf",
cannot open file 'Rplots.pdf'
Calls: boxplot -> boxplot.default -> do.call -> par -> <Anonymous>
Execution halted


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main.r
Run
Output
data <- c(200, 300, 400, 600, 1000)
minmax <- (data - min(data)) / (max(data) - min(data))
zscore <- (data - mean(data)) / sqrt(mean((data - mean(data))^2))
cat("Original Data:\n")
print(data)
cat("Min-Max Normalization:\n")
print(round(minmax, 3))
cat("Z-Score Normalization:\n")
print(round(zscore, 3))
Original Data:
[1] 200 300 400 600 1000
Min-Max Normalization:
[1] 0.000 0.125 0.250 0.500 1.000
Z-Score Normalization:
[1] -1.061 -0.707 -0.354 0.354 1.768


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Programiz
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main.r
Run
Output
age <- c(13, 15, 16, 16, 19, 20, 20, 21, 22,
22, 25, 25, 25, 25, 30, 33, 33, 35,
35, 35, 35, 36, 40, 45, 46, 52, 70)
mean_value <- mean(age)
median_value <- median(age)
freq <- table(age)
mode_value <- as.numeric(names(freq)[freq == max(freq)])
midrange <- (min(age) + max(age)) / 2
Q1 <- median(age[1:15])
Q3 <- median(age[15:27])
cat("Mean =", mean_value, "\n")
cat("Median =", median_value, "\n")
cat("Mode =", mode_value, "\n")
cat("Midrange =", midrange, "\n")
cat("Q1 =", Q1, "\n")
cat("Q3 =", Q3, "\n")
Mean = 29.96296
Median = 25
Mode = 25 35
Midrange = 41.5
Q1 = 20
Q3 = 35


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Programiz
R Online Compiler
Programiz PRO
main.r
Run
Output
lower <- c(1, 5, 15, 20, 50, 80)
upper <- c(5, 15, 20, 50, 80, 110)
freq <- c(200, 450, 300, 1500, 700, 44)
cf <- cumsum(freq)
N <- sum(freq)
half <- N / 2
i <- which(cf >= half)[1]
L <- lower[i]
f <- freq[i]
CF <- if (i == 1) 0 else cf[i - 1]
h <- upper[i] - lower[i]
median_value <- L + ((half - CF) / f) * h
cat("Total Frequency =", N, "\n")
cat("Approximate Median =", median_value, "\n")
Total Frequency = 3194
Approximate Median = 32.94


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main.r
Run
Output
speed <- c(78.3, 81.8, 82, 74.2, 83.4,
84.5, 82.9, 77.5, 80.9, 70.6)
cat("Mean =", mean(speed), "\n")
cat("Q1 =", quantile(speed, 0.25), "\n")
cat("Q3 =", quantile(speed, 0.75), "\n")
IQR_value <- IQR(speed)
cat("Interquartile Range =", IQR_value, "\n")
cat("Sample Standard Deviation =", sd(speed), "\n")
boxplot(speed,
main = "Boxplot of Car Speeds",
ylab = "Speed")
Mean = 79.63
Q1 = 77.7
Q3 = 82.675
Interquartile Range = 4.975
Sample Standard Deviation = 4.445835
ERROR!
Error in (function (file = if (onefile) "Rplots.pdf" else "Rplot%03d.pdf",
cannot open file 'Rplots.pdf'
Calls: boxplot -> boxplot.default -> do.call -> par -> <Anonymous>
Execution halted